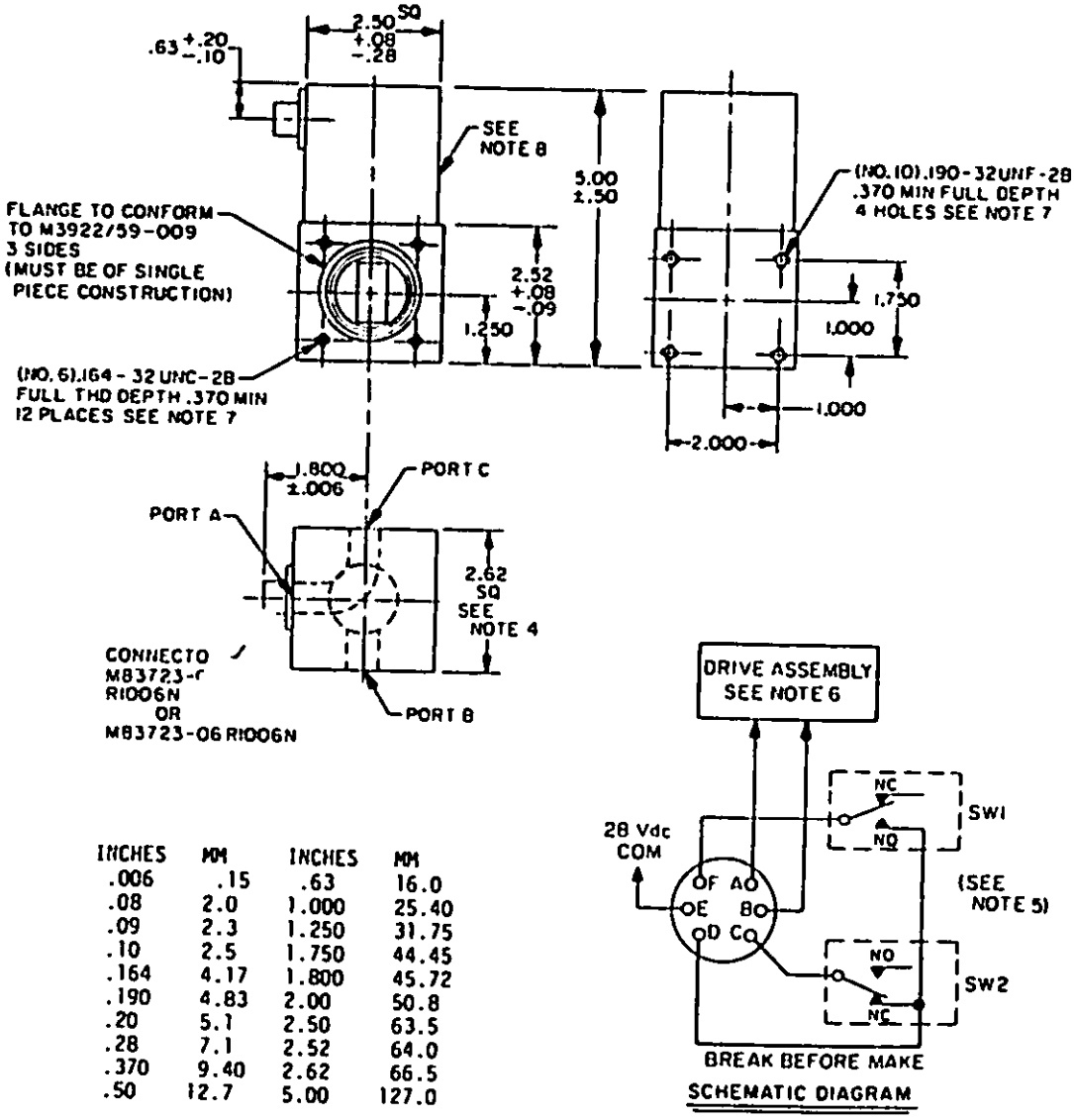
MILITARY SPECIFICATION SHEET

SWITCH, WAVEGUIDE, ELECTROMECHANICAL, LATCHING,
FREQUENCY RANGE 8.6 - 10.3 GHz

This specification sheet is approved for use by the Engineering and Technical Support Division (AFLC), Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the switch described herein shall consist of this document and the latest issue of Specification MIL-S-55041.



INCHES	MM	INCHES	MM
.006	.15	.63	16.0
.08	2.0	1.000	25.40
.09	2.3	1.250	31.75
.10	2.5	1.750	44.45
.164	4.17	1.800	45.72
.190	4.83	2.00	50.8
.20	5.1	2.50	63.5
.28	7.1	2.52	64.0
.370	9.40	2.62	66.5
.50	12.7	5.00	127.0

FIGURE 1. Switch configuration and schematic, part no. M55041/10-001.

FSC 5985

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.02 (.5 mm) for two place decimals and ±.005 (.13 mm) for three place decimals.
4. Rotor sequence: When the circuit A-E is energized, the RF energy path is from port A to C. When circuit B-E is energized, the RF energy path is from port A to B.
5. Interlock switch operation sequence: When circuit A-E is energized, the circuit C-D must open prior to circuit F-D being closed. When circuit B-E is energized, circuit F-D must open prior to circuit C-D being closed. All normally closed contacts and normally open contacts must open and close respectively within or before the first .03 (.8 mm) inch of rotor travel and within or after the last .03 (.8 mm) inch of rotor travel. The rotor shall come to a stop in its end position without visible bounce and shall not operate the interlock switch during such an oscillation.
6. Drive assembly operation: The switch shall not require any holding current. The drive assembly shall be automatically deenergized when the rotor is latched. The direction of drive assembly moving the rotor shall be as specified in the pictorial view.

   Case 1: After circuit A-E is energized, if the actuation voltage is removed from it and the rotor is stopped at some point after its first .03 (.8 mm) inch of travel and before its last .03 (.8 mm) inch of travel, the rotor shall continue to its new position (open energy path from port A to port C) by reapplying voltage to circuit A-E. However, the rotor shall return to its original position (open energy path from port A to port B) if the reapplied voltage is applied to circuit B-E.

   Case 2: After circuit B-E is energized, if the actuation voltage is removed from it and the rotor is stopped at some .03 (.8 mm) inch of travel, the rotor shall continue to its new position (open energy path from port A to port B) by reapplying voltage to circuit B-E. However, the rotor shall return to its original position (open energy path from port A to port C) if the reapplied voltage is applied to circuit A-E.
7. Helical thread inserts may be used.
8. The shape of the can is optional.

FIGURE 1. Switch configuration and schematic, part no. M55041/10-001 - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Exterior surface: Black finish in accordance with MIL-F-14072.

Weight: 2.5 (1.13 kg) pounds, maximum.

Operation: 1P2T latching, break before make.

Rotor operation: See figure 1, note 4.

Connector: The power connector shall be M83723-02R1006N (flange mount type) or M83723-06R1006N (jam nut type).

Drive assembly operation: See figure 1, note 6.

Drive assembly operating voltage: 21 Vdc to 29 Vdc.

Drive assembly starting current: 4 amperes, maximum, at 27 Vdc.

RF power handling capability: Peak 400 kW, average 400 W.

Temperature altitude: The temperature altitude shall be as specified for class 2 equipment as indicated in MIL-T-5422 except upper limit shall be 90°C in lieu of 71°C.

Frequency range: 8.6 - 10.3 GHz.

Isolation: 80 dB, minimum.

VSWR: 1.1:1, maximum.

Insertion loss: 0.10 dB, maximum.

Switching time (see figure 2): 1 second, maximum, 21 Vdc at 20°C ±10°C;
2 seconds, maximum, 21 Vdc at -54°C ±5°C.

Pressure: With 30 lb_f/in^2(gage) applied, the pressure drop shall not exceed 5 lb_f/in^2(gage) in 1 hour.

Pressure equilibrium: 1 lb_f/in^2. Tested as follows:

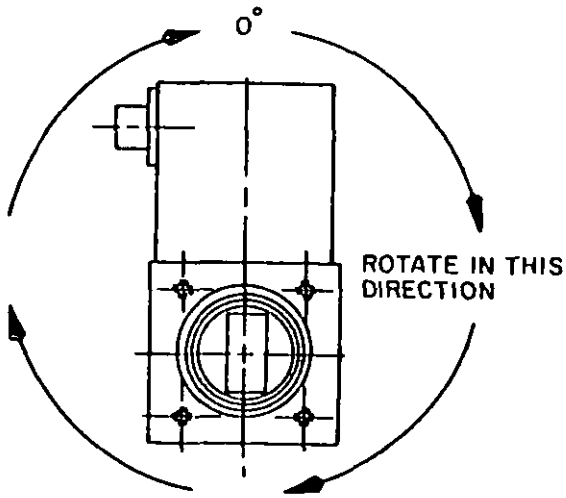
a. Seal one active (open) waveguide port by means of an O-ring and flat plate and attach a compressed air (15 lb_f/in^2(gage)) input to the other active (open waveguide port).
b. Adjust flow rate of compressed air to 100 cubic inches per minute.
c. Measure the pressure differential between the air input port to which the compressed air is attached and the inactive (closed) waveguide port. The pressure differential shall not exceed value specified.

Interlock switching current: 5 amperes, maximum, at 27 Vdc;
2 amperes, maximum, at 24 Vdc.

Cycle life: 50,000 cycles; rate of cycling shall be 10 per minute, maximum.

Part number: M55041/10-001.

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NOTE: Switching time: The switch shall be mounted in position shown. The switching time shall be taken and recorded. The switch shall be rotated in 45° increments and switching time recorded at each position thru 360°. The switching time shall not vary more than 10% from the time recorded at 0° (the original position).

FIGURE 2. Switching time.

NOTE: The switch covered by this specification sheet is substitutable for the following:

Manufacturer's code	P/N	NSN
97151	7334765	5985-00-179-4449

Custodian:
Air Force - 85

Review activity:
Air Force - 99

Preparing activity:
Air Force - 85

(Project 5985-F467)

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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